

Yifan Yuan

yifanyu2@andrew.cmu.edu | +1 (734) 353-3758

EDUCATIONAL BACKGROUND

The University of Melbourne

Melbourne, Australia

Bachelor of Science in Data Science; WAM: 87/100 (H1; GPA: 3.917/4.0; ~top 3% of cohort)

Dec. 2025

- **Awards:** Mathematics and Statistics Vacation Research Scholarship (2025); Global Scholarship (2024)

Exchange Program in University of Michigan, Ann Arbor

Ann Arbor, United States

College of Engineering; GPA: 3.97 / 4.0

Aug. 2024 – May. 2025

- **Key Modules:** Machine Learning (Graduate-level), Statistics, Stochastic Process, Optimization (Non-linear, Graduate-level)

Carnegie Mellon University, School of Computer Science

Pittsburgh, United States

Master of Science in Intelligent Information System

May. 2028

INTERNSHIP EXPERIENCES

Shanghai Electric

Shanghai, China

AI product development Intern

Feb 2026 - Jul. 2026

- Developed an end-to-end HR recruitment agent for complex candidate–role matching, combining structured competency analysis with iterative HR review for traceable screening.
- Translated ambiguous needs into an evolving talent intelligence network prototype, where existing employee profiles and potential candidate pools are continuously enriched and matched for emerging workforce needs to support internal-first mobility and external talent supplementation.
- Built a flexible-manufacturing industrial agent integrating equipment guidance, safety risk identification, and work-permit workflows, and showcased it as an applied AI case for AI adoption in manufacturing.

Schneider Electric

Shanghai, China

Machine learning Intern

May 2025 - Jul. 2025

- Initiated a multi-agent system for virtual factory operations and led company-wide workshops to support its adoption.
- Built a Python-based statistical modeling system with SQL integration for real-time production anomaly detection from 0 to 1, achieving 97.1% accuracy.
- Developed case-based AI vision tutorials for Lighthouse Factories, facilitating industry–academia integration.

Center for Healthcare Engineering & Patient Safety (CHEPS), University of Michigan

Ann Arbor, United States

Research Assistant, Reimagining Operations for Antenatal Care Delivery Team

Jan. 2025 – May. 2025

- Orchestrated a data-processing pipeline through case-study-based data cleaning, feature engineering, and cross-dataset integration to estimate tailored policy application in current healthcare system.
- Collaborated with clinicians to trace data back to clinical workflows, validating a reliable dataset for downstream analysis.
- Contributed to the methodology section of a clinical research paper and quantified operational and clinical impacts of policy.

Activ (Delphi)

Shanghai, China

Data Analysis Intern

Nov. 2023 - Feb. 2024

- Participated in the preliminary design of a big data platform and was responsible for its data processing, extracting and processing millions of daily records using SQL for analysis.
- Built interactive dashboards in Tableau to visualize dynamic data, such as energy consumption over normal time based on different subdivision levels and time windows.

RESEARCH EXPERIENCES

Buy-Now-Pay-Later (BNPL) Merchant Risk Modeling & Ranking System

Aug. 2025 - Nov. 2025

- Built an end-to-end ETL pipeline in Python consolidating multiple internal and external datasets into analysis-ready schema.
- Designed a custom KNN “look-alike” fraud model for extreme label sparsity (114 fraud merchants, ~13M transactions), optimizing feature weights to distinguish fraudulent merchants and propagate fraud risk.
- Engineered consumer loyalty, revenue concentration (HHI), and growth indicators to identify resilient, high-potential merchants.
- Prototyped an interactive merchant platform combining KNN fraud risk with customizable indicator-weighted rankings.

The Role of Multiplicative Noise in Cellular Decision-Making: Insights from SDEs

Jan. 2025 - Feb. 2025

- Applied stochastic differential equations (SDEs) in Julia to model noise-driven dynamics in complex systems, deriving steady-state probability distributions $P_s(x)$ and probabilistic landscapes $U_q(x)$.
- Measured how noise intensity shapes system behavior, such as driving transitions from monostable to bistable system.

Optimizing Image Classification: Layer Freezing, and LoRA Adaptation

Oct. 2024 - Dec. 2024

- Optimized layer-wise parameter freezing on ResNet-34, improving accuracy from 75.9% to 90.6% over FC-only fine-tuning while using 38% fewer parameters than full fine-tuning.
- Adapted LoRA from Transformers to CNN architectures, identifying architectural constraints in ResNet and reducing trainable parameters by 99.7% (119.6M $\rightarrow$ 0.33M) on VGG-16 FC layers with only a 3.5 pp accuracy decrease.

SKILLS

Programming: Python, R, C, Julia, VBA, SQL (DBeaver, MySQL Workbench)

Tools & Platforms: Excel, Tableau, PyTorch, PySpark, ProModel

Languages: Mandarin (Native), English (IELTS 7.5)